Python Activity-Based Learning
Program 6: Railway Fare and Reservation Calculator

Problem Statement

Calculate railway fares for multiple passengers using distance slabs, travel class, age concessions, reservation fees, and cancellation penalties.

Objectives

• Calculate fare based on distance and class.

• Apply age-based concessions.

• Calculate total reservation amount and cancellation refund.

Assumptions

• Base fare is ■0.50 per km for Sleeper and ■1.00 per km for AC.

• Reservation fee is ■40 per passenger.

• Children below 5 travel free; passengers aged 60 or above receive a 20% fare concession.

• Cancellation penalty is 25% of the fare before reservation fee.

Algorithm

1. Read passenger count and journey distance.

2. For each passenger, read name, age, and class.

3. Select the class rate and calculate base fare.

4. Apply age concession or free travel rule.

5. Add reservation fee where applicable.

6. Display passenger fare and cancellation refund.

Flowchart

```
START → Read distance and passenger count → Read passenger details → Select class rate → Calculate fare → Appl
```

Pseudocode

```
BEGIN
  READ distance, passenger_count
  FOR each passenger
    READ name, age, class
    IF class is Sleeper rate = 0.50 ELSE rate = 1.00
    fare = distance * rate
    IF age < 5 THEN fare = 0
    ELSE IF age >= 60 THEN fare = 80% of fare
    IF fare > 0 THEN total = fare + 40 ELSE total = 0
    refund = total - (fare * 25%)
    DISPLAY total and refund
  END FOR
END
```

Python Source Code

```
distance = float(input("Enter journey distance in km: "))
count = int(input("Enter number of passengers: "))

for i in range(count):
    print("\nPassenger", i + 1)
    name = input("Name: ")
    age = int(input("Age: "))
    travel_class = input("Class (sleeper/ac): ").lower()

    if travel_class == "sleeper":
        rate = 0.50
```

```
elif travel_class == "ac":
    rate = 1.00
else:
    print("Invalid class. Passenger skipped.")
    continue

fare = distance * rate

if age < 5:
    fare = 0
elif age >= 60:
    fare = fare * 0.80

if fare == 0:
    total = 0
    refund = 0
else:
    total = fare + 40
    refund = total - (fare * 0.25)

print("Passenger:", name)
print("Fare including reservation: Rs.", round(total, 2))
print("Refund after cancellation: Rs.", round(refund, 2))
```

Test Case

Distance 100 km; passenger 1: age 30, sleeper; passenger 2: age 65, AC.
Expected: Passenger 1 fare ■90 (■50 fare + ■40 fee), refund ■77.50. Passenger 2 fare ■120 (■100 less 20% + ■40 fee), refund ■95.

Sample Output

```
Passenger: Passenger 1
Fare including reservation: Rs. 90.0
Refund after cancellation: Rs. 77.5
Passenger: Passenger 2
Fare including reservation: Rs. 120.0
Refund after cancellation: Rs. 95.0
```

Conclusion

The program calculates passenger-wise railway fares, applies the specified age concessions, adds reservation fees, and estimates cancellation refunds.